

Prescient

Prescient 2026 Challenge Planning Handbook

A read-only challenge audit, validation design, modelling roadmap and Extra Trees decision guide

Prepared 5 September 2026 | No model was trained, fitted, tuned or backtested for this report

Audit basis	Finding
Repository	README, Python/R harnesses, validators, sample solutions and declared dependencies
Data	Schema and descriptive-quality inspection of both supplied CSV files only
Scope	Understand the problem and plan the work; no predictive modelling or performance claims
Recommended start	Simple rule-based signal or closed-form Ridge, with cost-aware smoothing
Extra Trees status	Optional challenger only; not the default and not yet operationally cleared

Repository safeguard: the challenge permits only `solution.py` or `solution.R` to change for submission. This handbook is a working document and must not be included in the pull request.

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Section	Purpose
1. Executive decision	What to optimize and what to build first
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Appendices	Formulas, audit facts, glossary and source trail

1. Executive decision

This is a **daily tactical asset-allocation** problem, not a security-selection problem. The system must choose long-only weights for six broad asset classes before seeing each day's return. It is paid only for net excess growth over a cost-free benchmark. The hidden 2026 year is half the score; the median of four stressed historical years is the other half.

Recommended starting architecture

Begin with a deliberately small signal set and one of two low-complexity engines: (a) a transparent regime/ranking rule, or (b) a closed-form Ridge regression. Convert expected relative returns into modest active tilts, legalize them with the supplied constraint function, and approach targets slowly or only when the benefit exceeds cost. This directly addresses the challenge's stated failure mode: curve-fitting.

Extra Trees verdict

Extra Trees is **nonlinear but not low-flexibility**. An ensemble can express many thresholds and interactions, even when each tree is depth-limited. It may be useful as a constrained challenger if nonlinear regimes are real, but current repository evidence does not justify making it the primary model. The dataset has only 5,500 dependent daily observations, the response is noisy, and the declared Python requirements do not include scikit-learn. Unless the grading environment explicitly guarantees that dependency, relying on sklearn would create a submission-failure risk; implementing Extra Trees from scratch inside the only permitted solution file would add complexity and runtime risk.

Decision in one sentence

Use simplicity as the default, then allow Extra Trees to displace Ridge only if a predeclared, leakage-safe test shows broad net improvement across development folds at matched active weight and turnover, while dependency, determinism and runtime gates all pass.

Success criteria

- First, pass every gate: unattended run, legal weights, deterministic output, average total active weight of at least 5% in every evaluation window, permitted inputs only, and total runtime under 10 minutes.
- Second, maximize net geometric daily excess return, not information ratio or raw forecast accuracy.
- Third, require regime breadth: positive or acceptably stable results across chronological development folds, not one attractive year.
- Fourth, keep the final system explainable enough to diagnose why a weight changed and why a trade was worth its cost.

2. Challenge objective

Plain-English statement

At the start of each trading day, choose what fraction of the fund to hold in South African equity, global equity in rand, South African bonds, South African cash, listed property and gold in rand. Only earlier observations are visible. Relative to a fixed balanced-fund benchmark, take a meaningful but constrained view, pay one-way trading costs for all weight changes, and beat the benchmark during an unseen 2026 period and several historical stress regimes.

Notation

Symbol	Meaning
t	Decision/evaluation trading day; the function sees data strictly before t
i	One of the six asset classes
$r(i,t)$	Asset i total return on day t , decimal form
$w(i,t)$	Portfolio weight selected before observing $r(i,t)$
$b(i)$	Fixed benchmark weight
$a(i,t)$	Active weight, $w(i,t) - b(i)$
$c(i)$	One-way trading-cost rate for asset i
$P(t)$	Net portfolio return after trading cost
$B(t)$	Benchmark return
$G(W)$	Scored daily geometric excess, in basis points, over window W

Daily returns and cost

$$B(t) = \sum_i b(i) r(i,t)$$
$$\text{Cost}(t) = \sum_i c(i) |w(i,t) - w(i,t-1)|$$
$$P(t) = \sum_i w(i,t) r(i,t) - \text{Cost}(t)$$

The harness starts each evaluation window holding the benchmark, so $w(i,t-1)$ equals $b(i)$ on the first day. Reported turnover is half the absolute two-sided weight change, but cost is charged on the full one-way amount by asset.

Scoring function

$$G(W) = 10,000 \times \{ [\text{product over } t \text{ in } W \text{ of } (1 + P(t))]^{(1/n)} - [\text{product over } t \text{ in } W \text{ of } (1 + B(t))]^{(1/n)} \}$$
$$\text{Score} = 0.50 \times G(\text{live } 2026) + 0.50 \times \text{median}\{G(2015), G(2016), G(2020), G(2022)\}$$

Important code-level nuance: the README describes mean daily net excess return, while the harness computes the **difference in geometric mean daily returns**. The harness implementation is the operational definition. Information ratio, tracking error and drawdown are reported diagnostics but do not enter the score.

Scale supplied by the organizer

- The sample solution is described as about +0.08 basis points per day.
- The README calls roughly +1 basis point per day genuinely good.
- Perfect foresight one week ahead is described as about +6 basis points per day.
- These are organizer guideposts, not results from this audit.

3. Rules, constraints and submission gates

Daily feasible set

Asset	Benchmark	Band-implied range	Effective notes	Cost (bps)
SA_EQUITY	40.0%	30.0%-50.0%	With global equity, total <=75%	15
GLOBAL_EQUITY	20.0%	10.0%-30.0%	ZAR-denominated total return	20
SA_BONDS	25.0%	15.0%-35.0%	ALBI total return	8
SA_CASH	7.5%	0.0%-17.5%	Long-only floor binds	1
SA_PROPERTY	5.0%	0.0%-15.0%	Highest trading cost	35
GOLD	2.5%	0.0%-10.0%	10% cap overrides 12.5% band top	25

For every day: $\sum_i w(i,t)=1$; $w(i,t) \geq 0$; $|a(i,t)| \leq 0.10$; $\sum_i |a(i,t)| \leq 0.40$;
 $w(\text{SA_EQUITY}, t) + w(\text{GLOBAL_EQUITY}, t) \leq 0.75$; and $w(\text{GOLD}, t) \leq 0.10$.

Because active weights net to zero in a fully invested portfolio, the L1 active budget of 40% means at most 20% of capital can be reallocated from underweights to overweights. Similarly, the minimum average total active weight of 5% means at least 2.5% of capital is reallocated on average.

Window-level gates

- Average total active weight must be at least 5% over each run. A benchmark-only solution fails.
- Runtime must be under 10 minutes. The Python validator times the practice window plus the four historical robustness windows; its duplicate determinism runs occur after the displayed timed block.
- Consecutive runs must be identical. The validator checks practice-window portfolio returns to a tolerance of $1e-12$.
- Execution must be unattended and must use only the supplied CSV files. No downloads, API calls, extra files or hard-coded market history are allowed.
- Submission must arrive before 14:00 on 5 September, as stated in the repository.

Illegal-day behavior

Local validation runs with strict constraint checking and fails the affected window. The README says the grading system instead substitutes benchmark weights for an illegal day, counts the violation, and charges the trade back toward the benchmark. Therefore the robust engineering target remains zero violations; relying on the fallback sacrifices the view and may incur two-way costs.

File and pull-request rules

- Choose Python or R. Only solution.py or solution.R may be changed for the submission; the organizer diffs the harness.
- The callable is `generate_weights(hist, prev_weights, params)`. It must return six weights as a dict, Series or array in asset order (named numeric vector in R).
- Declare every tunable number in PARAMS. The validator reports the count; the README says parameter count is part of assessment but gives no numeric penalty formula.
- The pull request should contain one changed solution file and disclose the team name, approach/why, and AI tools used.

Do not commit this handbook to the challenge pull request. Keep it as a local planning artifact or move it outside the repository before submission.

4. Data audit and data dictionary

Observed coverage and quality

Check	data_assets.csv	data_macro.csv
Rows	33,000 long-format rows	5,499 daily rows
Date range	5 Jan 2004 - 31 Dec 2025	5 Jan 2004 - 31 Dec 2025
Unique dates	5,500	5,499
Duplicates	0 date-asset duplicates	0 date duplicates
Missing cells	0	0
Cross-file alignment	All six assets on every asset date	No row on 23 Oct 2024
Frequency	Weekday/trading-day panel; 248-252 rows in most full years	Same calendar except one date

The missing macro date is not a null-filled row; it is absent. The harness passes macro rows with date strictly before the decision date and does not reindex them to asset dates. Feature code should therefore use a past-only as-of convention. Forward-filling from the latest earlier macro observation is permissible; backward-filling from a later date is leakage.

Asset file fields

Field	Type / unit	Intended use	Caution
datestamp	Date	Ordering and cutoffs	Never random-shuffle time
asset_code	Six categorical codes	Column identity	Preserve harness order
asset_name	Descriptive text	Documentation	Not a predictive feature
price	Positive total-return index level	Momentum, drawdown, ratios	Level is non-stationary; use returns/changes
return	Daily decimal total return	Targets, risk and momentum	0.01 means 1%; day t is unavailable at decision t

All six assets have 5,500 observations, no missing values and no non-positive prices. From the second date onward, price percentage changes reproduce the supplied return series to floating-point tolerance. The first return predates the first shown price and should not be reconstructed from the visible price file.

Asset definitions

Code	Meaning	Sample annualized volatility*	Portfolio role
SA_EQUITY	FTSE/JSE All Share Total Return	18.63%	Domestic growth/risk asset
GLOBAL_EQUITY	MSCI World Net Total Return in ZAR	16.39%	Foreign growth plus currency exposure
SA_BONDS	FTSE/JSE All Bond Index	7.57%	Duration/income and local-rate exposure
SA_CASH	STeFI Composite	0.30%	Low-volatility funding/defensive asset
SA_PROPERTY	FTSE/JSE SA Listed Property Total Return	17.17%	Rate-sensitive growth/income; costly to trade
GOLD	Gold spot per ounce in ZAR	20.64%	Commodity/currency/defensive diversifier

*Full-sample descriptive standard deviation $\times \sqrt{252}$, 2004-2025. These are data-audit statistics, not forecast or backtest results.

Macro file fields

Field	Interpretation / stored scale	Candidate transformation	Primary hypothesis
usdzar	Rand per US dollar, level	Log change; 21/63-day momentum; volatility	Higher values mean a weaker rand; may support ZAR global equity/gold
dxy	US dollar index, level	Log change; rolling z-score	Global dollar/risk/liquidity regime
vix	Equity implied-volatility index, level	Log level/change; past-only z-score	Risk-off state and nonlinear thresholds
brent	Brent oil price, level	Log return; trend; volatility	Inflation/commodity and SA terms-of-trade proxy
us_2y	US 2-year yield, percent	Level change; real-time z-score	Policy expectations
us_10y	US 10-year yield, percent	Change; us_10y-us_2y spread	Duration and global growth/inflation
sa_10y	SA 10-year yield, percent	Change; spread versus repo/JIBAR/US 10y	Local bond risk and risk premium
jibar_3m	3-month JIBAR, percent	Change; spread to repo/SA 10y	Money-market conditions
sa_repo	SA policy repo rate, percent	Change; time since change	Domestic monetary stance
em_equity	Emerging-market equity index, level	Log return; trend; drawdown	Global risk appetite relevant to SA assets

The README states that yields are stored in percent, not decimals. It does not document exact data vendors or quote currency for every macro series, so the implementation should avoid assumptions not established by the repository.

Useful descriptive relationships

Full-sample daily-return correlations show that SA equity and property are the strongest pair among the risk assets (0.455), global and SA equity correlate at 0.358, and gold is nearly uncorrelated with SA equity (-0.047). SA bonds are negatively correlated with global equity (-0.166) and gold (-0.221). These are unconditional historical summaries, not stable laws; regime variation is exactly what the challenge tests.

Data use map

Input	Use	Do not do
Past returns	Momentum, reversal, realized risk, correlations, targets whose full horizon has completed	Read return on the allocation day
Past prices	Equivalent multi-period returns and drawdowns	Feed raw trending levels without normalization
Past macro	Changes, spreads, regimes and conditional signals	Backward-fill missing dates or fit normalizers on future data
Previous weights	Turnover-aware adjustment and no-trade logic	Treat each day as costless independent optimization
Benchmark	Relative-return target and active-space allocation	Optimize absolute returns while ignoring benchmark exposure

5. Financial and statistical concepts

Active return and active weight

Active return is the portfolio's net return minus the benchmark return. Active weight is a position difference, not a risk measure: $A(t) = \sum_i |w(i,t) - b(i)|$. Two portfolios with the same active weight can have very different volatility because the assets have different risks and correlations.

Tracking error and information ratio

Tracking error is the annualized volatility of active returns. The optional `hist.te(w)` estimates ex-ante tracking error using a 250-day exponentially weighted covariance matrix with decay 0.97. Information ratio is average active return divided by active-return volatility, annualized. Both are diagnostic only; the challenge score is net excess growth.

Regimes and non-stationarity

A regime is a period in which relationships among growth, inflation, rates, currency and risky assets behave differently. The official years deliberately span a rand shock (2015), rebound (2016), pandemic shock (2020) and inflation shock (2022). A feature that worked on average across 2004-2025 can still fail when its sign or scale changes. Prefer transformations and models whose behavior is stable when levels shift.

Signal horizon versus trading horizon

Daily weights do not require a one-day prediction. A slower 5- to 20-day expected-return signal can be held and updated gradually while the harness still asks for weights each day. This is often better aligned with transaction costs and the low signal-to-noise ratio of daily asset returns.

Why cost awareness belongs inside the methodology

A forecast is economically useful only if its expected benefit exceeds the cost of moving from yesterday's portfolio. The cost hierarchy is extreme: property costs 35 bps per unit traded versus 1 bp for cash. Identical target changes should not automatically trigger identical trading behavior across assets.

Bias-variance and flexibility

A low-flexibility model may miss nonlinear regime effects but tends to be stable in a small, noisy sample. A high-flexibility model can fit thresholds and interactions yet mistake one historical episode for a durable rule. Extra Trees reduces the variance of a single tree by averaging randomized trees, but the ensemble remains capable of highly irregular functions unless depth and leaf size are constrained.

Forecast quality is not the final objective

- Mean-squared error can improve while portfolio score worsens if small forecasts cause excessive trading.
- Directional accuracy can be high on low-impact days and low when markets move most.
- A model can beat an absolute-return baseline yet fail against the fixed benchmark.
- Portfolio evaluation must match the exact harness, costs, constraints and starting weights.

6. Leakage-safe methodology

Information timing contract

For decision date t , `hist.returns` and `hist.prices` contain only dates earlier than t ; `hist.macro` also contains only dates earlier than t . The day's realized return is applied after `generate_weights` returns. This prevents direct same-day look-ahead inside normal use, but leakage can still enter through feature preprocessing, label construction, offline constants, validation design or stateful caches.

Recommended modelling target

Predict **relative** return over a short holding horizon rather than raw price levels. One option is $y(i,s,h)$ = cumulative return of asset i from $s+1$ through $s+h$ minus the corresponding benchmark cumulative return. At decision t , a training label is allowed only if its final realized date is earlier than t . For h -day targets, the most recent h observations are therefore unavailable for fitting.

Alternative low-variance targets

- Cross-sectional rank of future relative returns, which reduces sensitivity to extreme magnitudes.
- Risk-adjusted relative return with a predeclared volatility scaler, used only as a training target - not confused with the official score.
- A regime label (risk-on/risk-off, rising/falling local yields) followed by transparent allocation rules.
- A direct slow signal from past trends and macro changes, with no fitted predictive model at all.

Feature categories - design only

Category	Examples	Controls
Trend/momentum	5/21/63/126/252-day cumulative returns; fast-minus-slow	Use few horizons; avoid dense grids
Reversal	1- to 5-day return after extreme move	Small role; high turnover risk
Risk	Realized/downside volatility, drawdown, EWMA covariance	Past window only; robust clipping
Relative/cross-asset	Asset minus benchmark momentum; ranks; risk-asset breadth	Compute per date from available six assets
Currency	USDZAR trend/volatility; global-equity local-vs-currency context	No external foreign-currency series
Rates	US term spread; SA 10y-repo; changes and curve steepening	Yields are percentage points
Risk regime	VIX z-score/change; EM equity and DXY trend	Normalizer fitted expanding/rolling from past
Commodity	Brent and gold trend, volatility and divergence	Avoid contemporaneous target overlap
Portfolio state	Previous weight, distance to target, last trade size	Use to reduce churn, not predict returns

Feature hygiene

- Transform non-stationary levels into changes, spreads, rolling ranks or past-only z-scores.
- Shift every feature so its timestamp is no later than the most recent observation available before the decision.
- Fit winsorization, scaling and imputation separately inside each training origin; never on the whole dataset.
- Forward-fill macro values only from earlier dates and carry an optional staleness indicator.
- Avoid duplicate representations such as price returns and identical return-window features unless regularization is strong.

Leakage catalogue

Leakage path	Why it fails	Prevention
Random train/test split	Future regimes influence training and serial neighbors cross the boundary	Chronological expanding/rolling origins only
Whole-sample scaler	Future means, volatility and extremes enter earlier predictions	Fit transformer on each training slice
Overlapping target labels	Training labels can include returns inside the validation period	Purge h-1 boundary rows; embargo as needed
Same-day macro/return	The allocation is made before day t data exists	Features end at t-1 or latest earlier macro row
Hard-coded historical outcomes	Violates rules and cannot generalize to 2026	Algorithmic rules based solely on hist
Global cache across windows	Validator runs windows out of chronological order; future fitted state can bleed backward	Stateless functions or reset cache when dates/length move backward
Repeated official-window tuning	Converts the robustness years into a training set	Lock them as final challenge tests
Backfilled missing macro date	Uses 24 Oct information to fill 23 Oct	Past-only as-of/forward fill

Pooled versus per-asset modelling

Six separate models offer asset-specific behavior but each sees only one time series. A pooled panel creates up to 33,000 asset-date rows, but those rows are grouped into only 5,500 dates and are not independent. A prudent pooled model uses asset indicators and shared standardized features, groups validation by date, and interprets sample size in terms of dates rather than rows. Ridge benefits most clearly from pooling; Extra Trees may overstate confidence if the 33,000 rows are treated as independent.

Refit schedule

Do not refit a complex model every day. A monthly or quarterly refit schedule is enough for slow tactical signals and preserves runtime. Any in-memory cache must be deterministic and robust to the validator's sequence (2025, then 2015, 2016, 2020, 2022, then 2025 twice). A date moving backward must trigger a clean reset. A stateless closed-form model using an infrequent calendar condition is safer.

7. Validation plan

Validation hierarchy

Layer	Purpose	Data policy	Decision output
Unit tests	Check dates, shapes, shifts, legality and costs	Tiny hand-calculated slices	Correctness only
Development walk-forwards	Choose signal family and small hyperparameter set	Years outside official score windows	Model/feature decision
Stress slices	Diagnose crashes and turnover under shocks	Chronological development periods	Risk controls
Official local windows	Final challenge robustness check	2015, 2016, 2020, 2022, 2025	Pass/fail and last comparison
Live 2026	Hidden organizer evaluation	Unavailable by design	Final score

Protect the official windows

The repository encourages teams to run the four historical regimes, but repeatedly selecting features and parameters on those five visible evaluation years will overfit the exact test design. Recommended discipline: develop on chronological blocks drawn from 2004-2014, 2017-2019, 2021 and 2023-2024; reserve 2015, 2016, 2020, 2022 and 2025 for late-stage confirmation. Early years also supply warm-up history, so not every year should be a standalone fold.

Suggested rolling-origin structure

Origin example	Train information	Validation	Notes
A	2004 through 2009	2010	Early structural regime; warm-up available
B	2004 through 2010	2011-2012	Sovereign-risk and volatile period
C	2004 through 2012	2013-2014	Pre-official-window check
D	2004 through 2016*	2017-2019	Algorithm sees only past at each day; *official years are history by then
E	2004 through 2020*	2021	Post-shock transition
F	2004 through 2022*	2023-2024	Recent pre-practice environment

The asterisks recognize the live algorithm's expanding-history reality: later decisions legitimately include earlier official years as past observations. What remains prohibited is using a year's future returns to make decisions inside that same year or repeatedly choosing parameters because that year's final score was attractive.

Purging and embargo

For an h-trading-day target, remove training rows whose label window overlaps the first validation label window. Purge at least h-1 rows at each boundary. If features use slow market-state estimates, a short embargo can further reduce adjacency dependence, but it should be predeclared and not tuned against official outcomes.

Evaluation dashboard

Metric	Role	Minimum evidence sought
Official daily excess bps	Primary economic objective	Positive median across development folds
Worst-fold excess	Regime fragility	No catastrophic dependence on one regime
Cost drag	Implementation efficiency	Improvement survives net of asset-specific costs
Average turnover	Stability proxy	Explained by deliberate rebalance policy
Mean/peak active weight	Risk-budget use	Mean safely above 5%; peak <=40%
Violation count	Engineering quality	Exactly zero
Determinism	Gate	Exact repeatability to validator tolerance
Runtime	Gate	Comfortable margin below 600 seconds
IR / tracking error / drawdown	Diagnostics	No hidden risk explosion; not optimized directly
Forecast metrics	Model diagnosis	Secondary to portfolio economics

Model-selection rule

Select the smallest system on the performance-complexity frontier. A more complex candidate must improve median and lower-tail development-fold net excess at matched active weight and similar turnover. If gains disappear after turnover matching, the model is finding faster signals, not better economic signals. If gains depend on one fold, retain the simpler model.

Avoid false precision

With a few yearly folds and correlated daily returns, small score differences may be noise. Compare sign, fold breadth, rank stability, turnover-adjusted uplift and qualitative behavior during shocks. Do not use dozens of hyperparameter trials or declare a winner from a few hundredths of a basis point per day.

8. Baselines and staged model shortlist

Baseline ladder

Stage	Candidate	Why it exists	Expected disposition
0	Shipped inverse-volatility tilt	Verifies harness, constraints and cost plumbing	Feasibility reference only; README says it loses in practice
1	Static legal active tilt	Measures how little a fixed view contributes and verifies 5% gate	Diagnostic only; organizer warns fixed tilts earn near zero live
2	Slow momentum/risk rule	Tests whether timing with very few parameters adds value	Strong first real baseline
3	Rule-based macro regime overlay	Adds a small number of economically motivated nonlinear thresholds	Preferred simple nonlinear challenger
4	Closed-form Ridge	Combines correlated features with shrinkage and stable extrapolation	Recommended primary fitted model
5	Constrained Extra Trees	Tests whether threshold interactions add robust incremental value	Optional challenger only

Recommended simple baseline specification

- One medium-horizon relative-momentum measure per asset (for example, a predeclared blend of 3- and 12-month momentum, optionally skipping the latest week).
- One risk scaler based on recent volatility or covariance, used conservatively so cash does not mechanically dominate.
- One or two macro overlays with clear hypotheses, such as VIX risk regime and SA yield-curve direction.
- Cross-sectional demeaning/ranking so positive and negative tilts fund each other around the benchmark.
- Partial adjustment and asset-aware no-trade thresholds, with extra restraint for property and gold.

Closed-form Ridge design

Ridge solves $\beta = (X'X + \lambda I)^{-1}X'y$ after past-only standardization. It can be implemented with numpy, already permitted by requirements. A pooled model can share information across assets while using asset indicators or asset-specific interaction blocks. Refit monthly or quarterly, and map forecasts to weights through a separate, deterministic allocation layer.

Why a simple regime rule may beat a fitted model

The score rewards timing across a small number of broad assets, while the daily target is extremely noisy. A threshold such as high/rising VIX plus weakening EM equity may capture an economically meaningful state without estimating dozens of coefficients. Such a rule is easier to stress-test, cheaper to run and less likely to change drastically after one new observation.

9. Extra Trees suitability assessment

What Extra Trees actually is

Extremely Randomized Trees averages many decision trees. At each split it considers a random subset of features and randomized candidate thresholds, then chooses among those candidates. Averaging reduces variance relative to one tree, but the fitted function is still piecewise constant and can represent complex interactions. Calling it 'less flexible nonlinear' is only accurate if depth, leaf size and feature count are deliberately constrained.

Potential strengths here

- Can discover threshold behavior such as a momentum signal working differently when VIX or yields are high.
- Handles interactions without manually multiplying features and is insensitive to monotonic feature scaling.
- Averaging many randomized trees is usually more stable than a single decision tree.
- Prediction is fast once fitted, and six-asset portfolio decisions are small.
- Permutation-based diagnostics can reveal whether performance depends on one feature family, when computed strictly out of sample.

Material limitations in this challenge

- Only 5,500 dates exist. Pooling gives 33,000 asset-date rows, but six rows on a date share the same macro state and are not independent.
- Daily and even weekly future returns have weak signal and fat-tailed noise; randomized thresholds can chase rare crises.
- Trees do not extrapolate beyond training outcomes or feature ranges, a weakness for an unseen 2026 regime.
- Feature importance can be unstable or biased toward variables with many potential thresholds; permutation importance must respect time blocks.
- Unconstrained trees generate jumpy forecasts, which can translate into costly daily weight changes.
- Randomness must be fully seeded, and thread-level execution should be controlled for exact repeatability.
- scikit-learn is absent from requirements.txt. Because only the solution file may change, dependency availability is not guaranteed.
- A from-scratch forest inside solution.py is inconsistent with the goal of a simple, auditable three-hour solution.

Dataset interaction

Extra Trees becomes more plausible if the feature matrix is compact (roughly 10-30 stable features), targets use a slower horizon, fitting is pooled across assets, leaf sizes are large, and refits are infrequent. It becomes less plausible with hundreds of correlated lag features, one-day labels, fully grown trees, per-asset fitting, or daily retraining. The number of historical crisis episodes - not raw rows - is the binding sample-size constraint for regime discovery.

How to constrain Extra Trees if it is tested later

Control	Conservative starting principle	Reason
n_estimators	Enough for stable averages, not a large tuning grid	Reduces Monte Carlo variation; costs runtime
max_depth	Shallow, approximately 3-6	Limits high-order interactions
min_samples_leaf	Large, approximately 30-100 date-equivalent samples	Prevents crisis memorization
max_features	Fixed moderate fraction	Adds diversity without hiding all core signals
target horizon	Slow, cost-compatible horizon	Reduces label noise and forecast churn
refit	Monthly or quarterly	Controls runtime and instability
random_state	Single fixed integer	Determinism gate
n_jobs	Prefer 1 unless exact repeatability is proven	Avoid platform/thread variation
features	Same compact set used by Ridge	Fair incremental-nonlinearity test

These ranges are planning bounds, not selected hyperparameters. Any actual values must be declared in PARAMS and chosen on development folds, not official windows.

Decision framework

Use Extra Trees only if...	Do not use Extra Trees if...
The grader environment is confirmed to include a compatible sklearn without changing requirements	Dependency availability is uncertain or requires editing requirements.txt
A compact feature set has plausible threshold interactions	The thesis is mostly linear momentum/risk scaling
Blocked walk-forward improvement is broad and survives costs	Uplift comes from one year or disappears after turnover matching
Large leaves/shallow depth still improve lower-tail folds	Only deep/small-leaf trees perform well
Predictions and feature importance are stable across origins/seeds	Weight paths or importances change materially with small choices
Runtime has a large safety margin and repeated runs are exact	Fit time approaches the 10-minute gate or cache logic is fragile
The final explanation remains concise and economically defensible	The ensemble is used as a black box because it scores one split

Champion-challenger gate

Extra Trees may replace Ridge only when it improves both median and adverse-fold net excess on protected development origins, at comparable average active weight and turnover; remains helpful after removing any single feature family; produces zero violations; repeats exactly; and consumes comfortably less than half the runtime budget in local validation. Otherwise retain Ridge or the rule system.

Conceptual model comparison

Model	Flexibility	Strength here	Main weakness here	Current role
Rule-based	Very low	Transparent regimes, trivial runtime, direct cost control	May miss interactions; thresholds need discipline	Co-primary baseline
Ridge / linear	Low	Stable, fast, shrinkage, numpy implementation, extrapolates	Misses threshold interactions unless engineered	Recommended primary fitted model
Extra Trees	Medium-high even when averaged	Automatic nonlinearities and interactions	Small effective sample, jumpy outputs, sklearn dependency	Optional gated challenger
Random Forest	Medium-high	Less randomized, familiar robustness	More correlated trees; similar dependency/tuning issues	Lower priority than Extra Trees
Gradient Boosting	High	Can fit subtle tabular structure	Sequential overfit sensitivity and tuning burden	Not early-stage
XGBoost / LightGBM	High	Powerful regularized tabular learners	External packages, parameter burden, complexity	Avoid unless rules/environment clarified
ARIMA / VAR	Low-medium	Explicit time dynamics; VAR interpretable with few series	Return predictability weak; statsmodels absent; macro levels non-stationary	Small VAR/AR via numpy is optional
Neural network	Very high	Flexible shared representation	Too little independent data, unstable, opaque, dependency/runtime risk	Not recommended

Why Extra Trees is preferred over boosting only as a nonlinear test

If a tree ensemble is tried, Extra Trees offers a cleaner experimental question: do broad randomized threshold interactions add value beyond Ridge? Boosting layers many sequential decisions and usually invites a larger tuning search. Extra Trees can still overfit, but large leaves and shallow depth make its capacity easier to bound.

10. From signal to portfolio

Separate prediction from allocation

The predictor should output comparable scores or expected relative returns. A deterministic allocator then turns them into active tilts. This separation makes it possible to compare model families using the same risk budget, legality projection and trading policy.

Recommended allocation flow

1. Observe	2. Score	3. Normalize	4. Target	5. Trade	6. Legalize
Past-only returns, prices, macro, previous weights	Rule/Ridge (Extra Trees only if gated)	Cross-sectional demean, robust scale/rank	Benchmark plus modest active tilts	No-trade/partial adjustment by cost	Use <code>make_legal</code> ; verify independently

Target active risk

Do not automatically use the full 40% budget. The minimum average active weight is 5%, so a planning target around 8%-15% can provide a buffer without forcing extreme bets. The actual value must be validated later. Because the 5% check is window-specific, smoothing cannot leave the strategy near benchmark for long portions of a year.

Cost-aware trading options

- Partial adjustment: move a fixed fraction of the gap from previous to target weights.
- No-trade band: trade only when forecasted improvement or target gap exceeds an asset-specific threshold.
- Signal hysteresis: require stronger evidence to reverse a position than to maintain it.
- Scheduled rebalance: refresh slow signals weekly or monthly while still returning weights daily.
- Cost-weighted shrinkage: reduce tilt changes more aggressively for property and gold than for cash and bonds.

Constraint implementation

The supplied `make_legal()` works in active space and enforces long-only, bands, equity/gold caps, full investment and the 40% budget. Keep it unless tests reveal a defect. Still add unit checks around NaN/inf, exact asset ordering, sum-to-one tolerance, cap boundaries and pathological signals. Projection can alter the rank and size of raw model outputs, so evaluate the final legal portfolio, not only forecasts.

Numerical safety

- Return benchmark weights during warm-up or if a model fit is ill-conditioned - but ensure the yearly mean-active gate remains safely above 5% after warm-up.
- Use ridge regularization, pseudo-inverse/solve fallbacks and finite-value checks.
- Clip or rank extreme inputs using past-only limits.
- Avoid equality-edge targets; leave small buffers below 40% active, 75% equity and 10% gold caps.

11. Phased implementation plan

Phase 0 - freeze understanding (completed by this audit)

- Record repository rules, exact scoring code, asset order, costs, windows and dependency constraints.
- Confirm the repository is clean and preserve harness/data files unchanged.
- Treat the handbook as non-submission material.

Phase 1 - correctness scaffolding

- Create feature-timing unit tests and hand-check one decision date.
- Create a local evaluation summary that reproduces harness metrics without changing harness.py.
- Test make_legal on extreme, missing and reversed-rank signals.
- Build deterministic cache reset rules or choose a stateless implementation.

Phase 2 - simple real baselines

- Run the shipped sample solely to verify plumbing.
- Implement one slow momentum/risk rule with cost-aware smoothing.
- Add at most one macro regime overlay at a time, based on a written economic hypothesis.
- Choose active budget and trade speed using development folds only.

Phase 3 - Ridge champion

- Build a compact, versioned feature matrix with past-only transformations.
- Use a pooled or carefully regularized multi-output Ridge model.
- Refit monthly/quarterly using expanding history and purged targets.
- Compare against the rule baseline at matched turnover and active weight.

Phase 4 - nonlinear challenge

- First confirm package availability under the organizer's exact environment without modifying prohibited files.
- Test Extra Trees on the identical compact feature set, target and allocator.
- Use a tiny predeclared capacity grid; no broad automated search.
- Apply the champion-challenger decision gates in Section 9.

Phase 5 - lock and validate

- Freeze PARAMS and code before the official-window run.
- Run python validate.py (or R equivalent), then repeat in a fresh process.
- Check runtime margin, zero violations, average active weight, deterministic equality and only-one-file diff.
- Write the pull-request explanation and AI disclosure; do not include research artifacts.

Three-hour execution budget

Elapsed	Goal	Stop condition
0:00-0:20	Environment, clean diff, data load, exact sample harness run	Do not proceed until reproducible
0:20-0:55	Feature timing, legality and metric tests	No same-day access; all checks green
0:55-1:35	Simple momentum/risk baseline plus turnover control	One defensible baseline, no feature sprawl
1:35-2:10	Ridge challenger and development-fold comparison	Keep only broad, stable improvement
2:10-2:30	Optional constrained nonlinear test	Skip if dependency or runtime uncertain
2:30-2:50	Official validator, deterministic rerun, failure repair	All windows pass with margin
2:50-3:00	One-file diff, comments, PR narrative and AI disclosure	Submission package clean

Scope-control rule

If the simple baseline and Ridge are stable but Extra Trees is not clearly superior by 2:10, stop experimenting. The cost of an additional model is not just runtime; it is validation attention, parameter count, dependency risk and reduced ability to explain failures.

Parameter budget

Keep every tunable number in PARAMS and maintain a short rationale table in comments. Prefer one horizon, one risk lookback, one regularization strength, one active scale and one trade-control parameter. Adding a parameter requires a written hypothesis and broad fold evidence. The repository does not disclose a formal parameter-count penalty, so minimize count as a robustness principle.

12. Risks, failure modes and controls

Risk	Failure mode	Control
Curve-fitting	Good 2025/one regime, poor hidden 2026	Protected official windows, few features/parameters, worst-fold review
Hidden dependency	ImportError on grader	Use declared packages; confirm sklearn before Extra Trees
Look-ahead	Inflated history tests	Past-only feature pipeline, purged labels, no whole-sample transforms
Turnover	Gross signal consumed by costs	Slow horizon, no-trade bands, asset-specific restraint
Constraint edge	Illegal days and benchmark fallback	make_legal plus independent tolerance checks
Low activity	Mean active below 5%	Plan a safe buffer and monitor by window
Regime concentration	One crisis drives feature/model	Block diagnostics, leave-one-fold-out checks
State/cache contamination	Later-year model reused in earlier validator window	Stateless design or reset on backward date/length
Non-stationary macro levels	Spurious tree/linear splits	Changes, spreads, rolling ranks, past-only scaling
Outliers	Tree memorization or unstable regression	Robust transforms, large leaves, Ridge shrinkage
Runtime	Missed gate	Infrequent refit, vectorization, timing margin
Submission hygiene	Disqualification due to extra file	Final git diff must show one solution file only

Reproducibility and runtime checklist

- Pin the language path and use only packages present in requirements.txt (or base R).
- Sort all dates and preserve the six-asset order exactly.
- Use fixed numeric dtypes and explicit NaN/inf handling.
- Set all random seeds; prefer single-threaded stochastic models unless exact reproducibility is proven.
- Do not read any file other than the two supplied CSVs during grading.
- Avoid current time, environment-dependent paths, network access and nondeterministic iteration order.
- Reset caches when a new backtest starts or dates move backward.
- Time the full validator in a fresh process and retain at least a 2x safety margin where possible.
- Run the practice window twice and compare the full weight and return paths, not only the final score.
- Before submission, verify git diff/status and ensure only solution.py or solution.R changed.

13. Execution checklist for the next session

Start-of-session

- [] Confirm Python versus R; Python is the natural choice if Ridge is planned.
- [] Confirm repository status is clean and identify the exact solution file that will be edited.
- [] Record the current commit identifier and do not touch harness, validator, requirements or data files.
- [] Confirm whether scikit-learn is guaranteed in the grading runtime; if not, remove Extra Trees from the executable roadmap.

Correctness before modelling

- [] Write a feature availability assertion: max feature timestamp < decision date.
- [] Use past-only as-of handling for the missing macro row on 23 Oct 2024.
- [] Define the target horizon and ensure every training label ends before the fit date.
- [] Define cache reset behavior for out-of-order evaluation windows.
- [] Test all constraints and finite outputs on deliberately extreme signals.

Build sequence

- [] Establish the shipped sample and static-tilt diagnostic results without tuning.
- [] Implement one slow rule-based momentum/risk model and cost-aware adjustment.
- [] Evaluate on protected development walk-forwards; log net score, worst fold, turnover, active weight and violations.
- [] Add closed-form Ridge using the same features and allocator; refit infrequently.
- [] Only then, and only if dependency-cleared, run the constrained Extra Trees challenger.
- [] Select the least complex model that satisfies the predeclared robustness gate.

Finalization

- [] Freeze PARAMS before looking at official windows.
- [] Run all five visible windows with zero errors and zero illegal days.
- [] Verify each window's mean active weight is safely above 5%.
- [] Repeat from a fresh process and confirm identical outputs.
- [] Confirm runtime is comfortably under 600 seconds.
- [] Confirm the pull request shows exactly one changed solution file.
- [] Include team name, short method/why statement and AI-tool disclosure.

Recommended next-session decision

Implement the simple momentum/risk rule and Ridge first. Defer Extra Trees until package availability is confirmed and the linear baseline establishes what nonlinear uplift must beat.

Appendix A - audit facts at a glance

Item	Repository fact
Assets	6 broad asset classes
History	5 Jan 2004 through 31 Dec 2025
Asset observations	5,500 dates x 6 assets = 33,000 rows
Macro observations	5,499 dates x 10 variables; 23 Oct 2024 absent
Practice window	2 Jan 2025 - 31 Dec 2025, 250 trading days
Robustness windows	2015 (251), 2016 (249), 2020 (252), 2022 (251) trading days
Live window	Calendar 2026, hidden and merged by organizer
Benchmark	40% SA equity, 20% global equity, 25% SA bonds, 7.5% cash, 5% property, 2.5% gold
Main score	50% hidden live daily geometric excess + 50% median of four historical daily geometric excess scores
Allowed data	Only data_assets.csv and data_macro.csv
Allowed code change	Only solution.py or solution.R
Runtime	Under 10 minutes
Determinism	Required; validator repeats practice run
Minimum activity	Average total active weight $\geq 5\%$ per run

Audit boundary

This report inspected source files and computed descriptive schema/quality statistics only. It did not execute harness.py, validate.py, solution.py, or any alternative model. No model was fitted, tuned or backtested, and no performance result was generated.

Appendix B - compact glossary

Term	Definition
Active weight	Difference between portfolio and benchmark weight for an asset
Total active weight	Sum of absolute active weights; twice the capital reallocated when active weights net to zero
Basis point (bp)	0.01 percentage point; 1 bp = 0.0001 in decimal return
Benchmark	Fixed cost-free reference portfolio against which net performance is measured
Drawdown	Decline from the portfolio's prior cumulative-return peak
Embargo	Gap imposed near a validation boundary to reduce information dependence
Expanding window	Training set that grows through time while always ending before the decision
Geometric mean return	Constant daily growth rate that compounds to the observed terminal value
Information ratio	Annualized mean active return divided by active-return volatility
Leakage	Any use of information that would not have been available at the historical decision time
No-trade band	Threshold below which an intended weight change is suppressed to save costs
Purge	Removal of training observations whose future-label interval overlaps validation
Regime	A market state in which return, risk or cross-asset relationships differ
Ridge	Linear regression with L2 coefficient shrinkage
Tracking error	Volatility of portfolio returns relative to the benchmark
Turnover	Half the sum of absolute weight changes, as reported by the harness

Appendix C - source trail and interpretation notes

Repository materials inspected

- README.md - challenge narrative, data descriptions, rules, score, AI policy, setup and pull-request process.
- harness.py and harness.R - authoritative asset order, constants, History contract, constraints, costs, walk-forward mechanics and metric formulas.
- validate.py and validate.R - visible windows, runtime display, determinism check and pass/fail behavior.
- solution.py and solution.R - required callable, PARAMS convention, supplied inverse-volatility example and make_legal implementation.
- requirements.txt - declared Python dependencies: pandas, numpy, plotly and scipy.
- data/data_assets.csv and data/data_macro.csv - schema, coverage, duplicates, missingness, alignment and descriptive statistics.

Interpretation hierarchy

Where prose and code differ in precision, this handbook treats the harness implementation as operationally authoritative. The most important example is the daily excess metric: the code uses the difference between portfolio and benchmark geometric mean daily growth rates.

Known ambiguities to clarify with organizers if possible

- Whether the grading Python environment includes scikit-learn or only requirements.txt packages.
- Whether parameter count has a formal numeric penalty or is a qualitative review criterion.
- Whether the 10-minute gate is measured over the same set of runs and hardware as validate.py.
- The exact market-data providers/closing-time conventions for macro series, if intraday timing matters. The harness's strict previous-date interface should still be obeyed regardless.

Final planning conclusion

The challenge is best approached as robust, cost-aware tactical allocation under regime uncertainty. The winning complexity is likely to be lower than the largest model the data can technically fit. A compact rule or Ridge system is the rational default; a tightly constrained Extra Trees ensemble earns a place only as a controlled test of nonlinear interactions, not as an assumption.

End of handbook